

A categorical comparison of AU incompleteness, Beklemishev-Shamkanov, and Willard SJAS

R6 of the Refinement stage. Charter: [ADR-0004-lawvere-sjas-translation.md](#).

Status and scope. This is a comparison construction, not a claim that Beklemishev-Shamkanov's calculus or Willard's arithmetics are arithmetic universes. It identifies the categorical data used by the arithmetic-universe proof and then presents the other systems in the weakest common language that preserves their failure mechanisms. Claims about the three source corpora are page-image verified in [VERIFICATION.md](#).

0. Answer

The comparison needs **one more coordinate than the three Hilbert-Bernays-Löb (HBL) conditions**. For the usual formalized-G2 argument, the relevant package is

1. a selected Gödel/Löb fixed point $G \leftrightarrow \text{not } \Box G$;
2. the three HBL structure maps for the provability operation $\Box$; and
3. a diagonal on boxed hypotheses, $\text{copy}_A : \Box A \rightarrow \Box A \text{ tensor } \Box A$, expressing $\Box$ -contraction.

Willard's self-justifying construction uses a second, logically distinct diagonal construction: a sentence H that says that the theory obtained by adjoining H is consistent. Here H is the Group-3 axiom. The G2 fixed point G and the self-consistency fixed point H must not be identified merely because both are obtained by fixed-numeral substitution.

The initial arithmetic universe has the whole G2 package: finite products supply the diagonal, van Dijk-Oldenziel Propositions 6.5, 6.6 and 6.8 supply the HBL maps, and Lemma 6.12 supplies fixed points.

Beklemishev-Shamkanov's example S does **not** fail one of the three HBL conditions. It satisfies their versions L1-L3 and has every modalized fixed point needed here. What it lacks is the boxed diagonal. Categorically, its Lindenbaum category is symmetric monoidal rather than cartesian; $\Box A$ cannot in general be copied. Consequently the formalized-G2 arrow

$\Box(\Box \text{bottom} \multimap \text{bottom}) \multimap \Box \text{bottom}$

does not exist. B-S §6 also states that the empty-context sequent

$\Rightarrow \text{not } \Box \text{bottom}$

is not provable. Here $\Box\text{bottom}$ is the object-language assertion that contradiction is provable, and not $\Box A$ abbreviates $A \multimap \text{bottom}$; hence $\text{Con} := \text{not } \Box\text{bottom}$ is the Hilbert-style consistency formula for this provability modality. Categorically, a closed proof would be a morphism $I \multimap \text{Con}$, where I is the tensor unit representing the empty context. Thus S blocks the $G2$ derivation without proving its own consistency. The system is nevertheless externally nondegenerate: if it were inconsistent in B-S's Definition 2.1 sense, its consequence rules and cut admissibility would make the sequent of Proposition 4.1 provable.

Willard's Type-A $IS(A)$ lies on the other axis. Its ordinary classical logic has cartesian contraction, but its weak arithmetic cannot internalize the **uniform proof-code composition** required by HBL condition (2). In the syntactic proof doctrine there is no internal entailment

$\Box\Phi$ and $\Box(\Phi \multimap \Psi) \multimap \Box\Psi$

uniform in the two proof witnesses, although every fixed-proof-code instance can be verified. The same pointwise/uniform split supplies selected diagonal instances without proving total substitution: Willard 2001, Theorem A.1 footnote 16, gives the particular $G \multimap \text{not } \Box G$ needed by the HBL argument under that theorem's Π^1_1 -strength hypothesis, while the 1993 construction uses a different fixed instance to form Group-3. Group-3 is a global self-consistency point, and Willard separately proves external consistency. B-S's S also has external consistency; what it lacks is the **positive internal point**.

In short:

System	G2 fixed point	HBL maps	Boxed diagonal	Proves Con	External consistency
Joyal / van Dijk-Oldenziel AU	every $T : \Pi^1_1 \multimap \Pi^1_1$	all three	yes, cartesian	no point for the internal copy's consistency proposition if $U_\emptyset$ is consistent	assumed for $U_\emptyset$ in $G2$
Beklemishev-Shamkanov S	every modalized formula, including the needed G	all three	no	no	yes , as forced by Proposition 4.1
Willard Type-A $IS(A)$	selected G under Theorem A.1's strength hypothesis; Group-3 is separate	HBL (2) fails uniformly	yes	yes	yes , under Proposition 1's hypotheses (full)

This is the requested distinction: B-S exhibit an externally consistent system with **non-derivability of formalized G2**, but not a counterexample to non-formalized G2; Willard establishes a **consistent inhabitant of the self-consistency proposition**.

1. The common comparison object: a coded modal proof doctrine

Neither an arithmetic universe alone nor a bare modal algebra records all the distinctions at issue. We therefore use the following deliberately small interface. This is the Refinement's comparison construction, not terminology claimed from any of the three sources.

1.1 Structural category

Let K be a symmetric monoidal category of formula objects and derivations. Its tensor represents context combination, and its unit I represents the empty context. For a context $\Gamma = (A_1, \dots, A_n)$, write

$\text{ctx}(\Gamma) := A_1 \text{ tensor } \dots \text{ tensor } A_n$, with $\text{ctx}(\text{empty}) := I$.

An internal implication $A \multimap B$, when present, is characterized by the natural currying bijection

$K(\text{ctx}(\Gamma) \text{ tensor } A, B) \simeq K(\text{ctx}(\Gamma), A \multimap B)$.

Equivalently, in the associated closed symmetric multicategory, the same law is written

$K(\Gamma, A ; B) \simeq K(\Gamma ; A \multimap B)$,

where the entries before the semicolon are inputs and the entry after it is the output. Thus the left side consists of derivations of B from Γ, A , while the right side consists of derivations of $A \multimap B$ from Γ .

The notation $A \multimap B$ below is schematic at exactly one important point. When the implication is representable, as in the B-S consequence category, it is an object satisfying this adjunction. In the AU presentation it instead denotes van Dijk-Oldenziel's implication **between judgements**, interpreted in the appropriate classifying AU (Definitions 6.1-6.2). No exponential object in the base AU is being assumed. The theorem below needs the associated introduction, evaluation, composition, and contraposition steps, not cartesian closure.

Write not $A := A \multimap \text{bottom}$ throughout this comparison.

If tensor is a categorical product, every object has a diagonal

$\delta_A : A \rightarrow A \text{ tensor } A$.

Proof-theoretically, precomposition with δ_A is contraction. In a merely monoidal category no such diagonal is automatic.

1.2 Provability data

For each coded sentence A , let Prf_A be its object or predicate of proof witnesses, and let

$\Box A := \text{exists } p. \text{Prf}_A(p).$

The three HBL conditions become the following structure.

Name	Categorical/proof-doctrine form	Classical condition
Nec	global proofs of A lift to global proofs of $\Box A$	if $\vdash A$, then $\vdash \Box A$
BoxMP	$m_{A,B} : \Box(A \multimap B)$ tensor $\Box A \multimap \Box B$	$\Box(A \multimap B) \multimap (\Box A \multimap \Box B)$
Four	$\text{four}_A : \Box A \multimap \Box \Box A$	$\Box A \multimap \Box \Box A$

Across nonclassical logics, these three maps are not enough. The additional structural datum isolated by Beklemishev-Shamkanov is

$\text{BoxContr}_A : \Box A \multimap \Box A \text{ tensor } \Box A.$

In a cartesian doctrine this is just $\delta_{\Box A}$ and is invisible as a separate assumption. In an affine, linear, or other substructural doctrine it is real extra structure.

1.3 Two fixed-point roles, and consistency points

A fixed point is an object or proposition X equipped with derivations both ways between X and $E(X)$. Three strengths must not be conflated:

- **universal**: every endomorphism in a stated class has a fixed point;
- **modalized**: every syntactically modalized formula has a selected fixed point;
- **selected**: one specified endomorphism has a fixed point.

Two selected endomorphisms occur in this comparison:

- the **G2 diagonal** selects $G \multimap \text{not } \Box G$; this is the fixed point consumed by Theorem 1.1;
- a **self-consistency diagonal** selects a sentence H with $H \multimap \text{Con}(B + H, D)$, where the proof predicate itself depends on the theory obtained by adjoining H to the non-self-referential base B .

The first is an ingredient of the limitative argument. The second is how a self-referential consistency assertion is formed; adjoining H then turns it into a consistency point. Either selected instance can exist without an internally total substitution operation, and neither instance by itself implies the other.

For the chosen provability operation $\Box$, its internal Hilbert-style consistency proposition is

$\text{Con} := \text{not } \Box \text{bottom}$, that is, $\text{Con} := \Box \text{bottom} \multimap \text{bottom}.$

Because I is the tensor unit representing the empty context, a morphism out of I is a closed derivation. A *consistency point* is therefore a global derivation

$\text{con} : I \rightarrow \text{Con}.$

This is only the internal half of self-justification. The other half is the external, metatheoretic assertion that the theory is consistent. An inconsistent theory can of course inhabit every proposition, including Con .

1.4 The comparison theorem

Theorem 1.1 (categorical HBL-G2 package). Suppose the doctrine has a selected fixed point $G \leftrightarrow \text{not } \Box G$, Nec , BoxMP , Four , the instance BoxContr_G , and enough implication rules to perform the usual propositional packaging. Then the formalized-G2 arrow $\Box \text{Con} \rightarrow \Box \text{bottom}$ is derivable; under external consistency there is no consistency point.

Proof. Let $G \leftrightarrow \text{not } \Box G$ be the fixed point. Apply Nec and BoxMP to the forward direction $G \rightarrow (\Box G \rightarrow \text{bottom})$. This gives

$\Box G \rightarrow \Box(\Box G \rightarrow \text{bottom}).$

Four gives $\Box G \rightarrow \Box \Box G$. To apply BoxMP once more, now with $A = \Box G$ and $B = \text{bottom}$, the same input $\Box G$ must feed both arrows. This is the one critical use of

$\text{BoxContr}_G : \Box G \rightarrow \Box G \text{ tensor } \Box G.$

The resulting composite is $\Box G \rightarrow \Box \text{bottom}$. Hence

$\text{Con} \rightarrow \text{not } \Box G$, and therefore $\text{Con} \rightarrow G$ by the reverse fixed-point arrow. If we apply Nec and BoxMP to $\text{Con} \rightarrow G$, we get $\Box \text{Con} \rightarrow \Box G$; composition therefore gives the advertised $\Box \text{Con} \rightarrow \Box \text{bottom}$. If Con had a global proof, Nec would supply $\Box \text{Con}$, formalized G2 would supply $\Box \text{bottom}$, and Con itself would yield bottom . This is the ordinary HBL proof with the normally implicit cartesian diagonal made explicit. Beklemishev-Shamkanov Proposition 3.8 identifies the same contraction use in their abstract proof. ■

The theorem is a **sufficient package**, not a biconditional. The displayed proof consumes only the named instances—in particular BoxContr_G , not a diagonal at every object. If a needed instance is unavailable, this derivation is blocked; mere absence of the general family does not by itself prove that the conclusion is underivable, manufacture a consistency point, or prove external consistency.

2. Joyal / van Dijk-Oldenziel: the full cartesian package

2.1 Their actual AU construction

Let U_0 be the initial arithmetic universe and U'_0 the externalization of its internal initial arithmetic-universe object. Van Dijk-Oldenziel construct

- an interpretation functor $R : U_0 \rightarrow U'_0$;
- a global-sections functor $\Gamma : U'_0 \rightarrow U_0$; and
- the provability operation $\Box := \Gamma \circ R$.

A proposition is a monomorphism $\phi \rightarrow 1$. Because an arithmetic universe is a list-arithmetic pretopos, its structural tensor is a finite product. Every proposition and every boxed proposition therefore has a diagonal.

The paper stresses that an arithmetic universe need not be cartesian closed. Section 6 consequently treats $\phi \vdash \psi$ as an inclusion of subobjects and interprets implications between such judgements in a classifying arithmetic universe. The comparison here follows that construction; it does not pretend that $\mathcal{U}_0$ has an ordinary exponential implication.

2.2 HBL as AU arrows

The source's propositions instantiate the common interface exactly as follows.

Common datum	van Dijk-Oldenziel	Construction
Nec	Proposition 6.5	$\mathcal{U} \models \phi$ implies $\mathcal{U} \models \Box \phi$, from the unit-like map $\phi \rightarrow \Gamma R(\phi)$ of Lemma 5.15
Four	Proposition 6.6	$\mathcal{U} \models \Box \phi \vdash \Box \Box \phi$, by applying $\Box = \Gamma R$
BoxMP	Proposition 6.8	internal evaluation $\text{Hom}(\phi', \psi') \times \text{Hom}(1', \phi') \rightarrow \text{Hom}(1', \psi')$ in the classifying AU
ordinary MP	Propositions 6.7, 6.9, 6.10	composition and packaging of judgements
fixed point	Lemma 6.12	every $T : P1' \rightarrow P1'$ has a global fixed point
contraction	finite products in the AU	$\text{delta}_X : X \rightarrow X \times X$

Thus Theorem 6.13 derives Löb in §6, which the authors explicitly introduce as a sketch. The direct G2 result is Theorem 5.20; equivalently, the false/consistency instance of the Löb package rules out an internal proof of the consistency of $\mathcal{U}'_0$ when $\mathcal{U}_0$ is consistent.

The load-bearing map for comparison with the other systems is Proposition 6.8's evaluation. It is a single internal, uniform arrow. The existence of its value at every externally named pair would be weaker.

3. Beklemishev-Shamkanov: HBL without the boxed diagonal

3.1 The APS as a thin category

Beklemishev-Shamkanov Definition 2.1 starts with a preorder

$S = (L_S, \leq_S, \text{top}, \text{bottom})$.

View it as a thin category $L(S)$: its objects are sentences (or their deductive-equivalence classes), and there is a unique arrow $A \rightarrow B$ exactly when $A \leq_S B$. Definition 2.3 then adds the object operations $\Box$ (provability) and $\boxtimes$ (refutability), subject to APS conditions C1-C4. This is already an order-enriched categorical presentation of their abstract provability structure.

3.2 The consequence calculus as a monoidal category

Definitions 3.1-3.2 refine the preorder to multiset contexts. After quotienting by deductive equivalence, set

$K_S(A, B) = \{*\}$ if $A \vdash_S B$, and empty otherwise.

The thin syntactic category obtained by quotienting formulas by mutual derivability will be called the Lindenbaum category; compare Lawvere (1969/2006, §3). For its monoidal and symmetric-monoidal-closed forms, see Szabo (1978, Chapters 2-3 and 7-8).

Multiset union is represented by their conjunction tensor; top is the unit; and I_3 gives the closure adjunction on the representing context object

$K_S(\text{ctx}(\Gamma) \text{ tensor } A, B) \sim K_S(\text{ctx}(\Gamma), A \rightarrow B)$.

Before replacing the context by its representing object, this is simply I_3 's multicategorical sequent law

$\Gamma, A \vdash_S B$ if and only if $\Gamma \vdash_S A \rightarrow B$.

Exchange is built in because contexts are multisets. Cut is categorical composition: for the concrete system S , Theorem 5 proves cut admissible, so this presentation really is compositional. Lemma 5.1 also proves weakening admissible, so its monoidal unit is terminal: categorically the concrete system is symmetric affine monoidal closed. What is absent is neither composition nor weakening, but the diagonal that would make the tensor cartesian.

Their three Löb conditions are precisely:

- L1: $\Box(A \rightarrow B) \rightarrow (\Box A \rightarrow \Box B)$ (BoxMP);
- L2: $\Box A \rightarrow \Box \Box A$ (Four);
- L3: global proofs lift under $\Box$ (Nec).

The modal rule of the concrete calculus supplies these conditions. Its fixed-point constructor supplies $\text{fp } x.A \sim A(\text{fp } x.A)$ for every formula modalized in x , so in particular it supplies the $G \leftrightarrow \text{not } \Box G$ instance used in Theorem 1.1. Proposition 4.3 also gives infinitely many inequivalent Gödelian and Henkinian fixed points. Here *Gödelian* is B-S's Definition 2.5 term for the Jeroslow-shaped equation $p \leftrightarrow \boxtimes p$; it should not be confused with the particular unprovability fixed point denoted G in this comparison.

3.3 The precise missing morphism

Contraction would be a family

$\text{delta_A} : A \rightarrow A \text{ tensor } A.$

Restricted $\Box$ -contraction is only the family

$\text{delta}^\Box_A : \Box A \rightarrow \Box A \text{ tensor } \Box A.$

Beklemishev-Shamkanov Remark 3.9 shows that this restricted family, together with correspondingly weakened APS assumptions, is already enough for their G2 theorem. Their concrete system deliberately has no such family. In sequent form, one cannot uniformly pass from

$\Gamma, \Box A, \Box A \vdash B$

to

$\Gamma, \Box A \vdash B.$

This is why the concrete S can satisfy L1-L3 and have fixed points while Proposition 4.1 states that the formalized-G2 arrow is absent (the paper calls this easy from cut-freeness and does not display a proof):

$\Box(\Box\text{bottom} \rightarrow \text{bottom}) \not\rightarrow \Box\text{bottom}.$

This also explains why an affine surface syntax is not enough: B-S note that an affine Peano arithmetic can still admit $\Box$ -contraction. The categorical test is the existence of $\text{delta}^\Box$, not the absence of an unrestricted contraction rule.

3.4 Why this is not self-justification

Their Section 6 says explicitly that the example is not a counterexample to non-formalized G2, because

$\Rightarrow \text{not } \Box\text{bottom}$

is not provable. Categorically, $K_S(I, \text{Con})$ is empty.

This is the appropriate internal consistency formula for their modality. In the implication setting B-S define $\Box A := \Box \text{not } A$ and observe that $\Box\text{top} =_S \Box\text{bottom}$ whenever L1 holds. Thus $\text{not } \Box\text{bottom}$ negates their object-language inconsistency assertion; it is not the external claim that S is consistent.

This does **not** mean that S is externally inconsistent. In fact, the result of Proposition 4.1 forces consistency in the paper's Definition 2.1 sense. If $\text{top} \leq_S \text{bottom}$, condition I4 would give $\vdash_S \text{bottom}$. For the antecedent $F := \Box(\Box\text{bottom} \rightarrow \text{bottom})$, the bottom initial sequent gives $F, \text{bottom} \vdash_S \Box\text{bottom}$; I2—implemented for the concrete calculus by Theorem 5's admissible cut—would then give $F \vdash_S \Box\text{bottom}$, contrary to Proposition 4.1. The example therefore occupies the cell

fixed points + HBL + no BoxContr + no Con point + external consistency.

B-S gives a non-cartesian counterexample showing that the HBL conditions L1-L3 alone do not entail formalized G2. The additional structural datum used by the usual proof is contraction of the relevant boxed hypothesis. It provides a consistent system that does not assert its own consistency, not a self-justifying system.

4. Willard: cartesian logic without internal uniform proof composition

4.1 Two levels that must be kept separate

Willard's generalized arithmetic is a pair (α, D) of an axiom basis and a deductive apparatus. Its categorical presentation must distinguish:

1. the external category in which mathematicians can manipulate finite proof codes; and
2. the internal syntactic category of operations and entailments whose totality the weak theory α itself can prove.

Let C_α be the cartesian syntactic category of α : objects are definable contexts and arrows are equivalence classes of functional relations whose totality and functionality α proves. Its predicate doctrine

$$P_\alpha : C_\alpha^{\text{op}} \rightarrow \text{Pos}$$

sends a context to its preorder of formulas modulo provable equivalence; reindexing is substitution. Finite meets and existential quantification give the usual first-order hyperdoctrine structure. The proof relation is a predicate

$$\text{Prf}_{\{\alpha, D\}} \text{ in } P_\alpha(\text{Sent} \times N).$$

For a fixed sentence Φ , pull it back along its code to obtain the proof predicate P_Φ over N , and put

$$\Box_D \Phi := \text{exists } p. \text{Prf}_{\{\alpha, D\}}(\text{code}(\Phi), p) \text{ in } P_\alpha(1).$$

The terminal fiber $P_\alpha(1)$ is a genuine thin category: an arrow is an implication proved by α , and proved implications compose. This repairs the earlier draft's invalid "wide class of affordable arrows", which was expressly not closed under composition and hence was not a category.

The ordinary classical first-order logic of $IS(A)$ supplies cartesian context management, so contraction and the diagonals δ_X are present. Willard does not use the B-S escape hatch.

4.2 HBL (2) as uniform evaluation

For fixed sentences Φ, Ψ , HBL (2) has the following proof-witness form in the fiber over $N \times N$:

$$\begin{aligned} &\text{Prf}(\text{code}(\Phi), x) \text{ and } \text{Prf}(\text{code}(\Phi \rightarrow \Psi), y) \\ &\rightarrow \text{exists } z. \text{Prf}(\text{code}(\Psi), z). \end{aligned}$$

Its universal closure is equivalent, by ordinary first-order quantifier rules, to the terminal-fiber BoxMP entailment

$$\Box \Phi \text{ and } \Box(\Phi \rightarrow \Psi) \rightarrow \Box \Psi.$$

A stronger but familiar witness for this entailment would be a proof-combiner map

$\text{comp_Phi,Psi} : P_Phi \times P_(\text{Phi} \rightarrow \text{Psi}) \rightarrow P_Psi.$

Van Dijk-Oldenziel Proposition 6.8 obtains exactly the analogous uniform operation from evaluation on internal Hom objects.

Willard's 1993 technical report, printed p. 12, identifies the breach for Type-A IS(A): the system can prove the required composition statement only when the proof-code variables x and y are fixed constants. In categorical terms, pulling the witness entailment back along a named point

$\langle x\text{-bar}, y\text{-bar} \rangle : 1 \rightarrow N \times N$

can yield a proved instance, while there is no entailment uniformly over the whole product. "Every named instance" does not assemble into the uniform evaluation morphism. This is analogous to, but not an identification with, the local-versus-global existence warning in van Dijk-Oldenziel Remark 5.12.

The missing internal arrow is not the external nonexistence of proof composition. A metatheorist can combine tableaux derivations and appeal to cut elimination; the failure is that the weak arithmetic does not prove one uniform, total proof-code relation over variable x and y . The source's page-12 passage identifies that failure. The surrounding design—weak arithmetic together with a cut-free apparatus whose composition is supplied metatheoretically by cut elimination—explains why fixed inputs and a uniform internal transformation come apart. This comparison does not claim an independent complexity lower-bound theorem.

Accordingly, for the Type-A tableaux line the documented failure is the uniform witness entailment, and hence

BoxMP_Phi,Psi (HBL condition (2)),

not contraction. This document does not claim that every Willard configuration fails the same condition. The source corpus identifies condition (2) for IS(A); for the separate Hilbert/no-growth line, the exact member of the HBL triple is not pinned down in the audited sources. External consistency plus a self-consistency theorem forces some premise of the applicable G2 package to fail, but it does not identify which one.

4.3 Two selected diagonals without total substitution

Willard uses two selected fixed points here, and the comparison must keep them separate.

The fixed point used by G2. Willard 2001, Theorem A.1 assumes that α proves all of PA's $\text{Pi}^{\text{-}}_1$ theorems. Its footnote 16 defines

$\text{Theta}(z) := \text{forall } x,y. (\text{Subst}(z,x) \rightarrow \text{not Derive}(x,y))$

and lets N be the code of Theta . It then states that α proves, for $G := \text{Theta}(N)$,

$G \leftrightarrow \text{not Der}(\text{code}(G)).$

This is exactly the selected $G \leftrightarrow \text{not } \Box G$ consumed by Theorem 1.1. Theorem A.1 is stated-only in the Codification—the paper expressly omits a formal proof—but footnote 16 displays this fixed-point step and says where its $\text{Pi}^{\text{-}}_1$ -strength hypothesis is used.

The fixed point used by Group-3. Write $B = \text{UNION}(A)$ for the non-self-referential Group-1 and Group-2 base, and let

$\text{Subst}_i \rightarrow N \times N$

be the definable relation saying that the second code results from substituting the first code into itself. The 1993 technical report, printed p. 37, states that $\text{IS}(A)$ is too weak to prove

$\text{forall } x. \text{ exists } y. \text{ Subst}_i(x, y),$

but for every fixed numeral $k\text{-bar}$ it can prove

$\text{exists } y. \text{ Subst}_i(k\text{-bar}, y).$

Categorically, the projection $\text{Subst}_i \rightarrow N$ is not proved internally to be a cover and has no proved global choice map, while the named fiber used in Equations (A.2)-(A.3) is inhabited. Those equations define a different selected fixed point $H = \text{SelfRef}(B, D)$ satisfying

$H \leftrightarrow \text{not exists } p. \text{ Prf}_{\{B + H, D\}}(\text{code}(\text{bottom}), p).$

This second fixed point forms the self-referential Group-3 axiom. Neither selected construction is the universal fixed-point property of van Dijk-Oldenziel Lemma 6.12, and neither proves total substitution. The Theorem A.1 strength hypothesis scopes the first construction as used here; the Group-3 construction and Proposition 1 have their own, different hypotheses.

4.4 The positive data that make an SJAS

Let $H := \text{SelfRef}(B, D)$ and

$\alpha^* := B + H.$

Because H is constructed to express $\text{Con}(\alpha^*, D)$ and is an axiom of α^* , that consistency proposition has a global proof in the syntactic doctrine:

$\text{self} : 1 \rightarrow \text{Con}(\alpha^*, D).$

That alone would be cheap: an inconsistent theory has such a point too. Willard's Proposition 1 supplies the other half—under its stated hypotheses on the base and tableaux apparatus, α^* is externally consistent. The Codification records its proof status as `full`.

For a Type-A $\text{IS}(A)$ in the overlap of the stated hypotheses, the resulting categorical profile is therefore

selected G2 FP + selected Group-3 FP + cartesian contraction + Con point + external consistency + no uniform BoxMP.

The HBL/Löb composite cannot turn `self` into a contradiction because its evaluation leg is not an internal morphism. B-S's S is nondegenerate too; the strictly stronger positive fact on Willard's side is that Willard supplies the desired consistency point inside that nondegenerate theory.

For later Willard systems using SelfCons_0 or SelfCons_1, replace Con by the precisely coded Level-0 or Level-1 consistency predicate. One must not move between those predicates without rechecking the corresponding proof relation and consistency theorem.

5. The comparison square

The systems differ on two independent axes.

	Structural diagonal on boxed hypotheses	Uniform internal HBL evaluation
AU	present automatically ($x \rightarrow \langle x, x \rangle$)	present by Proposition 6.8
B-S S	absent ($\Box A \rightarrow \Box A$ tensor $\Box A$)	present as L1
Willard Type-A IS(A)	present in ordinary classical logic	absent internally ; fixed proof-code instances only

The fixed-point axis is separate again:

	Strength of diagonalization
AU	every endomorphism $T : P1' \rightarrow P1'$ (Lemma 6.12)
B-S S	every syntactically modalized fixed-point expression, including $G \leftrightarrow \text{not } \Box G$
Willard IS(A)	selected $G \leftrightarrow \text{not } \Box G$ under Theorem A.1's strength hypothesis; a distinct selected Group-3 self-consistency fixed point; no internally proved total substitution

Finally, self-justification adds positive and semantic data:

	$1 \rightarrow \text{Con}$	external consistency
AU with full package	ruled out when consistent	hypothesis of G2
B-S S	absent by their §6	yes , as forced by Proposition 4.1 and the consequence rules
Willard IS(A)	present as Group-3	proved under Proposition 1's hypotheses (full)

These tables should not be collapsed into a single “weakness” ordering. B-S weaken structural logic while keeping the modal derivability rules. Willard keeps structural logic and weakens what arithmetic can certify uniformly about its proof predicate. The systems meet only at the consequence: one leg of the G2 composite is missing. Both examples are externally nondegenerate; only Willard also supplies the internal consistency point.

6. What has and has not been proved

Proved in this comparison. From the source definitions, the three presentations instantiate the common interface as tabulated; the missing morphism in each case is identified; and Theorem 1.1 explains why each missing leg blocks the applicable HBL-G2 derivation. Proposition 4.1 together with the B-S consequence rules entails external consistency of S. The B-S and Willard failure points are not identified with one another, nor are Willard’s G2 and Group-3 fixed points.

Cited rather than re-proved. B-S cut admissibility, Proposition 4.1’s proof-omitted failure of formalized G2, and non-provability of not $\Box\text{bottom}$; Willard’s consistency-preservation theorems, selected fixed-point constructions, and exact Type-A HBL (2) failure; and the AU construction and its explicitly sketch-level §6 Löb derivation. Their proof status and image anchors are recorded in [VERIFICATION.md](#).

Not claimed.

- $\text{IS}(A)$ is not shown to be an arithmetic universe or a sub-AU of $\mathcal{U}_0$.
- The B-S monoidal category is not cartesian, and completing it cartesianly would add the very contraction whose absence is the theorem.
- The assignment $A \mapsto \Box A$ need not be an endofunctor in Willard’s internal doctrine; failure to transport arrows uniformly is the point.
- Unavailability of a required HBL or contraction instance blocks this G2 derivation, but is never by itself sufficient for self-justification or a general proof that the conclusion is underivable.
- The G2 fixed point and Group-3 self-consistency fixed point are not the same proposition, and their source hypotheses are not interchangeable.
- Naming HBL (2) as a failed Type-A condition does not assert that every instance of Nec and Four holds for every possible choice of the inner basis A; no uniqueness-of-breach claim is needed here.
- The exact HBL breach is not asserted uniformly across every Willard deduction method.
- **Pakhomov’s $H_{\{\omega\}}$ is not in the comparison.** *Scope note added 2026-09-04 during the Refinement’s adversarial review.* R1–R3 establish it as one of only **two** demonstrated routes to a self-verifying theory — it proves $\text{Con}^{\text{pred}}(H_{\{\omega\}})$, and like Willard’s systems it cannot prove successor totality — so its absence from §5’s square is a gap, not a neutral choice. §0’s closing contrast (“Willard establishes a consistent inhabitant of the self-consistency proposition”) should be read as a statement about the three systems compared here, not about the field. What a fourth row would need is a categorical presentation of a *set* theory whose witness is a finite model on the superexponential cut; no Lawvere-side treatment of it is held, and this comparison does not attempt one. See [refined-sjas.md](#) §1 and [R3-the-margin.md](#) §5.1.

7. Primary-source anchors

- F. William Lawvere, *Diagonal Arguments and Cartesian Closed Categories*, Lecture Notes in Mathematics 92 (1969), pp. 134-145; reprinted in *Reprints in Theory and Applications of Categories* 15 (2006), pp. 1-13: §3, especially reprint p. 10, for the term and construction of a Lindenbaum category.

- M. E. Szabo, *Algebra of Proofs*, Studies in Logic and the Foundations of Mathematics 88, North-Holland, 1978: Chapters 2-3 and 7-8 for monoidal, symmetric monoidal, monoidal closed, and symmetric monoidal closed syntactic categories.
- J. van Dijk and A. Gietelink Oldenziel, *Gödel incompleteness through Arithmetic Universes after A. Joyal*, arXiv:2004.10482: Lemma 5.15; Theorem 5.20; §6 Definitions 6.1-6.3, Propositions 6.5-6.10, Lemma 6.12, Theorem 6.13, Remark 6.14. Local witness: [AU source PDF](#).
- L. Beklemishev and D. Shamkanov, *Some abstract versions of Gödel's second incompleteness theorem based on non-classical logics*, arXiv:1602.05728: Definitions 2.1, 2.3, 3.1-3.4, 3.7; Proposition 3.8; Remark 3.9; Propositions 4.1 and 4.3; Lemma 5.1; Theorem 5; §6. External consistency of S is the comparison's consequence of Definition 2.1, I4, Proposition 4.1 and Theorem 5, rather than a quoted source theorem. Local witness: [B-S source PDF](#).
- D. E. Willard, *Self-Verifying Axiom Systems and the Incompleteness Theorem*, TR 93-10 (the collated witness cannot predate April 1994): Proposition 1, printed p. 8 and §§4-6 (full); printed p. 12 (fixed-witness-only HBL composition); printed pp. 37-38 (fixed-instance substitution and Equations (A.2)-(A.3) for Group-3). Local witness: [1993 technical report PDF](#).
- D. E. Willard, *Self-verifying axiom systems, the incompleteness theorem and related reflection principles*, JSL 66 (2001): Theorem A.1 (stated-only) and its footnote 16 selected fixed-point justification, pp. 44-45; consistency results as indexed in the Codification. Local witness: [2001 paper PDF](#).
- Corpus extraction and proof-status records: [1993 extraction](#), [2001 extraction](#), and the [result registry](#).